

Summary of *General Agents Contain World Models*

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The follow-up to *Robust Agents Learn Causal World Models*: that paper asked what an agent must know to stay near-optimal when the *environment* shifts; this one asks what it must know to pursue many *goals* in a fixed environment. Again competence forces a model—now predictive rather than causal—with accuracy forced up by the *depth* of the goals achieved.

Basic setup

The environment is a *controlled Markov process* (cMP): an MDP with no reward and no discount, given by a tuple $(\mathcal{S}, \mathcal{A}, P)$ with finite state space $\mathcal{S}$, action space $\mathcal{A}$, and transition function

$$P_{ss'}(a) = P(S_{t+1} = s' \mid S_t = s, A_t = a).$$

Assumption 1. The cMP is finite, stationary (transition probabilities do not change over time), *communicating* (every state is reachable from every other under some policy), has at least two actions, and is fully observed.

A *world model* is an approximation $\hat{P}$ of the transition function with bounded error, $|\hat{P}_{ss'}(a) - P_{ss'}(a)| \leq \varepsilon$ for all s, s', a . Rolled forward, such a one-step predictor predicts the environment under any policy.

An *agent* is a deterministic goal-conditioned policy, a map $\pi(h_t, \psi) = a_t$ from the history h_t and the goal ψ to an action. Two objects must be kept apart: $P_{ss'}(a)$ is the *environment's* transition law and is never an input to the policy. The theorem is striking precisely because P can nonetheless be reconstructed from the policy's behaviour. The paper's question is whether an agent that pursues a rich family of goals must, in this sense, determine a model of P .

Goals

Goals are built in three layers, from linear temporal logic.

Base goals. A base goal is $\varphi = \mathcal{O}([(s, a) \in \mathbf{g}])$, where $\mathbf{g}$ is a set of goal state–action pairs and $\mathcal{O}$ is one of three temporal operators,

$$\top = \text{now}, \quad \bigcirc = \text{next transition}, \quad \diamond = \text{eventually}.$$

These differ in horizon and in *what they constrain*. *Now* and *next* look alike but do different jobs: the atomic predicate is on a state–action pair, so $\top[(s, a)]$ says “be in s and take a at the current step”—it pins down the agent's own action, which it controls—whereas $\bigcirc[(s', \cdot)]$ says “be in s' at the next step”—it pins down the environment's response, which it does not. The two-stage goal $\langle \top[(s, a)], \bigcirc[(s', \cdot)] \rangle$, “do a in s and land in s' ”, is exactly the unit trial the extraction repeats: without *now* a goal could not force a particular action, without *next* it could not refer to the outcome. *Eventually* is different again: “eventually visit a state in $\mathbf{g}$ ” can range over an arbitrarily long trajectory. Long-horizon *structure*, as opposed to long duration, comes from sequential composition.

Sequential goals. $\psi = \langle \varphi_1, \dots, \varphi_n \rangle$ requires the sub-goals to be achieved in order, φ_i before φ_{i+1} . Its *depth* is $\text{depth}(\psi) = n$: the number of sequentially composed base goals. This is *not* the number of environment time steps—a single $\diamond$ sub-goal may take arbitrarily many transitions—but the number of stages the agent must chain.

Composite goals. $\psi = \bigvee_{i=1}^m \psi_i$ is a disjunction of sequential goals; its depth is the largest depth among them. Ψ_n denotes all composite goals of depth at most n .

Write $\tau \models \psi$ when trajectory τ satisfies ψ ; the success probability of a policy from s_0 is $P(\tau \models \psi \mid \pi, s_0)$. Two features of this goal language do all the work later: depth n measures how many stages a task chains, and a disjunction offers the agent *alternatives it must choose between*, which is what makes its preferences observable. These are not arbitrary preferences: choosing between incompatible branches reveals which has the higher *achievable* success probability, which is why behaviour carries numerical information about the dynamics.

Agents and regret

An *optimal* goal-conditioned agent maximises success on every goal from every start state, $\pi^* = \arg \max_{\pi} P(\tau \models \psi \mid \pi, s_0)$ for all $\psi \in \Psi$ and all s_0 with $P(s_0) > 0$.

The object of study is weaker, and it is the paper’s organising definition. A *bounded goal-conditioned agent* with maximum goal depth n and failure rate $\delta \in [0, 1]$ is any policy satisfying

$$P(\tau \models \psi \mid \pi, s_0) \geq (1 - \delta) \max_{\pi'} P(\tau \models \psi \mid \pi', s_0) \quad \text{for all } \psi \in \Psi_n.$$

In words: *competence* is how close the agent’s probability of satisfying each goal comes to the best achievable, uniformly over every goal in Ψ_n . This is an external performance criterion—the agent need not know the optimum, or that one exists—and it is the only assumption on the agent. It is the counterpart of δ -optimality across domain shifts, with goals playing the role interventions played; the regret is *multiplicative* (a fraction of the optimum) rather than additive.

Main theorems

Theorem 1 (world models from bounded regret). Let the environment satisfy Assumption 1 and let π be a bounded goal-conditioned agent with failure rate δ on all goals in Ψ_n , with $n > 1$. Then π fully determines a model $\hat{P}$ with

$$|\hat{P}_{ss'}(a) - P_{ss'}(a)| \leq \sqrt{\frac{2P_{ss'}(a)(1 - P_{ss'}(a))}{(n-1)(1-\delta)}} \sim O\left(\frac{\delta}{\sqrt{n}}\right) + O\left(\frac{1}{n}\right) \quad (\delta \ll 1, n \gg 1).$$

bounded regret at depth $n \implies$ policy determines a world model to $O(\delta/\sqrt{n}) + O(1/n)$.

Three readings. The error vanishes as the agent improves ($\delta \rightarrow 0$) *or* as its goals deepen ($n \rightarrow \infty$); any fixed $\delta < 1$ is overcome by large enough n , so the policy of even a mediocre agent that chains long tasks determines an increasingly accurate model. The bound is absolute, so rare transitions ($P_{ss'}(a) \ll 1$) may be poorly resolved relatively.

Theorem 2 (myopic agents are exempt). Even an optimal agent on *every* one-step goal need not determine any nontrivial approximation to P : for the paper’s myopic goal class the only bound on P derivable from π^* is the trivial $\varepsilon = 1$, and this is tight: competence on one-step problems reveals only the comparisons needed to choose actions, not the numerical transition probabilities.

So goal-directedness alone forces nothing; it is *depth*, the need to chain stages, that forces a model. Two qualifications belong right here. First, Theorem 1 establishes *recoverability*: an approximate transition model can be extracted from the policy by an external observer. It does not establish that the agent explicitly represents $\hat{P}$ or computes with it. Second, the converse is the familiar direction—given P , or a good enough $\hat{P}$, goal-optimal policies follow by planning—and it is in this combined sense that the paper says a world model “cannot be avoided”: the information is provably present in the behaviour, whatever the internal mechanism.

Extraction from the policy

As before, the mechanism turns the agent’s *choices* into numbers, and the parallel is exact once seen. Offer a composite goal $\psi_a \vee \psi_b$ whose branches cannot both be pursued: an optimal agent commits to

the branch with the larger maximal success probability, a δ -bounded one to a branch within a factor $(1 - \delta)$ of the best. Each query is thus a slightly noisy comparison of two success probabilities, read from a single action.

Measuring one transition. To measure $p := P_{ss'}(a)$, pose a goal that makes the agent run n independent trials of the same sub-task: go to s , take a , observe whether the outcome is s' , and return to s (the cMP is communicating). The number of trials that land on s' is

$$K \sim \text{Binomial}(n, p).$$

Now compare two incompatible sub-goals parameterised by a threshold r : $\psi_{\leq r}$, “at most r of the n trials succeed”, and $\psi_{\geq r+1}$, “at least $r + 1$ succeed”. Offered the choice, the agent prefers whichever is likelier. As r sweeps from 0 to n the preferred branch flips exactly once, at the threshold r_* where the two are balanced,

$$P(K \leq r_*) \approx P(K \geq r_* + 1) \iff r_* \text{ is the median of } \text{Binomial}(n, p),$$

and the binomial median lies within one of the mean np , so the observed switch gives

$$p \approx \frac{r_*}{n} \quad \text{up to } O(1/n).$$

Regret blurs the switch: a δ -bounded agent may take a branch merely within a factor $(1 - \delta)$ of the best, smearing the flip over the window of r where the two probabilities are that close; binomial concentration (standard deviation $\sqrt{np(1-p)}$) makes that window $O(\delta\sqrt{n})$ in r , i.e. $O(\delta/\sqrt{n})$ in p . The two effects add to exactly Theorem 1’s bound. This is why depth helps: n is both the goal’s depth and the number of trials, and more trials sharpen the binomial and the switch. It is the role the *continuum* of mixing weights q played before—resolution came from sweeping q continuously; here it comes from repeating a sub-goal n times—and a myopic agent ($n = 1$) has one coin flip and no resolution, which is Theorem 2.

$\text{choice between branches} \rightarrow \text{switching threshold } r_* \rightarrow \text{binomial median} \rightarrow P_{ss'}(a) \rightarrow \hat{P}.$

The resulting procedure (the paper’s Algorithm 1) is unsupervised and universal: its only input is π , and it works for every agent and environment meeting the assumptions.

Relation to the previous paper

The two results are halves of one picture. *Robust Agents* concerns *domain* generalisation: adapting to many interventions forces a *causal* model (G, P) . This paper concerns *task* generalisation: achieving many goals in a fixed environment forces a *predictive* model $P_{ss'}(a)$. The authors’ striking corollary: domain generalisation needs *strictly more* knowledge than task generalisation, because a transition function is observational and causal structure—how the environment responds to intervention—is not identifiable from it. A task-general agent need not know the causal graph; a domain-general one must.

Interpretation and consequences

With the recoverability reading fixed above, four consequences follow. *Safety*: a checkable world model can be elicited from any sufficiently capable model-free agent, as many verification and safe-exploration proposals presuppose. *Capability bounds*: building a general agent is at least as hard as learning an accurate world model, so regret-bounded generality is confined to domains with learnable dynamics. *Emergence*: a mechanism for implicit world models in foundation models, since minimising regret across diverse tasks forces one. And a motivation for model-free methods falls away: the model is learned either way.

Two limits for discussion. The result assumes a fully observed, stationary Markov environment, a deterministic policy, and a regret bound uniform over *every* goal up to depth n ; what a partially observed agent must learn about latent state is open; and Theorem 2 draws the boundary sharply—only multi-stage structure forces the model.